$$\left.\begin{aligned} 2x+5y &= 19 \\ 3x+4y &= 18 \end{aligned}\right\} \quad \underbrace{\begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}}_{A} \begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} 19 \\ 18 \end{bmatrix}}_{B}$$

$$\begin{aligned} 2x + \tfrac{8}{3}y &= 12 \\ - \qquad -3y &= -12 \\ \hline 0 + \left(5 - \tfrac{8}{3}\right)y &= 7 \end{aligned}$$

$$\left[\begin{array}{c:c} A & B \end{array}\right]$$

Augmented matrix

$$\frac{7}{3}y = 7, \quad y = 3$$

$$2x = 19 - 15 = 4, \quad x = 2$$

$$\left[\begin{array}{cc:c} 2 & 5 & 19 \\ 3 & 4 & 18 \end{array}\right] \begin{matrix} R_1^0 \\ R_2^0 \end{matrix} \xrightarrow[R_2^1 = R_2^0 - \frac{3}{2}R_1^0]{R_1^1 = R_1^0} \left[\begin{array}{cc:c} 2 & 5 & 19 \\ 0 & 4-\frac{15}{2} & 18-\frac{3}{2}19 \end{array}\right]$$

$$\left[\begin{array}{cc:c} 2 & 5 & 19 \\ 0 & -\frac{7}{2} & -\frac{21}{2} \end{array}\right] \xrightarrow[R_2^2 = \frac{R_2^1}{A_{22}}]{R_1^2 = R_1^1} \left[\begin{array}{cc:c} 2 & 5 & 19 \\ 0 & 1 & 3 \end{array}\right] \quad y = 3$$

$$2x + 15 = 19$$

$$x = 2$$

$$\left.\begin{aligned} x+y+z&=6\\ 2x+3y+z&=13\\ x+2y+3z&=13 \end{aligned}\right\} \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6\\ 2 & 3 & 1 & 13\\ 1 & 2 & 3 & 13 \end{array}\right] \begin{array}{l} R_1^0\\ R_2^0\\ R_3^0 \end{array} \xrightarrow[R_2^1=R_2^0-\frac{2}{1}R_1^0]{R_1^1=R_1^0}$$

$$R_3^1=R_3^0-\frac{1}{1}R_1^0$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6\\ 0 & 1 & -1 & 1\\ 0 & 1 & 2 & 7 \end{array}\right] \xrightarrow[R_2^2=R_2^1]{R_1^2=R_1^1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6\\ 0 & 1 & -1 & 1\\ 0 & 0 & 3 & 6 \end{array}\right]$$

$$R_3^2=R_3^1-\frac{1}{1}R_2^1$$

$$\downarrow$$

$$R_3^3=R_3^2/A_{33}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6\\ 0 & 1 & -1 & 1\\ 0 & 0 & 1 & 2 \end{array}\right] \Rightarrow \begin{array}{l} x=1\\ y=3\\ z=2 \end{array}$$

Gauss-Jordan Elimination

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6\\ 2 & 3 & 1 & 13\\ 1 & 2 & 3 & 13 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6\\ 0 & 1 & -1 & 1\\ 0 & 1 & 2 & 7 \end{array}\right]$$

$$\downarrow$$

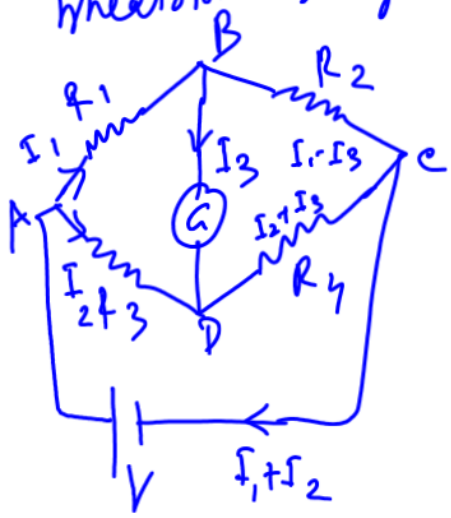
$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1\\ 0 & 1 & 0 & 3\\ 0 & 0 & 3 & 6 \end{array}\right] \longleftarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 5\\ 0 & 1 & -1 & 1\\ 0 & 0 & 3 & 6 \end{array}\right]$$

R_1^3/A_{11}

R_3^3/A_{33}

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1\\ 0 & 1 & 0 & 3\\ 0 & 0 & 1 & 2 \end{array}\right] \Rightarrow x=1,\ y=3,\ z=2$$

Wheatstone bridge Circuit



ADBA loop

$$I_2R_3 - I_3G - I_1R_1 = 0$$

$$R_1I_1 - R_3I_2 + GI_3 = 0 \quad \cdots (1)$$

DCBD loop

$$R_4(I_2+I_3) - R_2(I_1-I_3) + GI_3 = 0$$

$$R_2I_1 - R_4I_2 - (R_2+R_4+G)I_3 = 0 \quad \cdots (2)$$

ABCA loop

$$R_1I_1 + R_2(I_1-I_3) = V$$

$$(R_1+R_2)I_1 - R_2I_3 = V \quad \cdots (3)$$

$$\left[\begin{array}{ccc|c} R_1 & -R_3 & G & 0 \\ R_2 & -R_4 & -(R_2+R_4+G) & 0 \\ R_1+R_2 & 0 & -R_2 & V \end{array}\right]$$

The upward velocity of a rocket is given at three different times as

Time (sec)	velocity m/s
5	110
10	170
15	345

If $v(t) = at^2 + bt + c$

$5 \le t \le 15$

Find a, b, c.

$a = 2.2999$
$b = -22.5$
$c = 165.0$

$t=5;\quad 110 = 25a + 5b + c$

$t=10;\quad 170 = 100a + 10b + c$

$t=15;\quad 345 = 225a + 15b + c$

$$\left[\begin{array}{ccc|c} 25 & 5 & 1 & 110 \\ 100 & 10 & 1 & 170 \\ 225 & 15 & 1 & 345 \end{array}\right]$$